(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.JAGARAPU RAMESH**

Registration Number: **324129510024**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

ANOMALY DETECTION SYSTEM FOR IOT SENSOR NETWORKS USING MACHINE LEARNING AND REAL-TIME DATA MONITORING

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs.K.Pushpalatha

Department of CSE

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.JAGARAPU RAMESH

Reg.No: 324129510024

Department of CSE

Department of Computer Science and Engineering

Wellfare Institute of Science, Technology and Management

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Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or industry.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship program, you are the ambassador of your College, and your behavior during the internship program is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship program immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **Mr.JAGARAPU RAMESH**, a student of **Bachelor of Technology** Program, Reg. No. **324129510024** of the Department of **Computer Science and Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs.K.Pushpalatha**, Department of **Computer Science and Engineering**, **Wellfare Institute of Science, Technology and Management**.



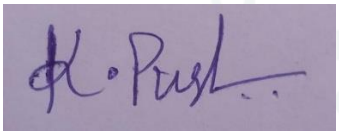
(Signature and Date)

Official Certification

This is to certify that **Mr.JAGARAPU RAMESH** , Reg. No. **324129510024** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **ANOMALY DETECTION SYSTEM FOR IOT SENSOR NETWORKS USING MACHINE LEARNING AND REAL-TIME DATA MONITORING** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Computer Science and Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements



Faculty Guide



Head of the Department

Head Dept of CSE
WISTM Engg. College
Pinagadi, VSP



Principal

Certificate from Intern Organization

This is to certify that **Mr.JAGARAPU RAMESH**, Reg. No. **324129510024** of **Wellfare Institute of Science, Technology and Management**, underwent internship in **ANOMALY DETECTION SYSTEM FOR IOT SENSOR NETWORKS USING MACHINE LEARNING AND REAL-TIME DATA MONITORING** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mrs.K.Pushpalatha** , Head of the Department of **Computer Science and Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr. G.MANIKANTA**, Assistant Professor of the Department of **Computer Science and Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this internship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

1.1 Introduction

This internship report details the design, development, and implementation of an Anomaly Detection System for IoT Sensor Networks using machine learning and real-time data monitoring. The project addresses the critical challenge of identifying abnormal sensor behavior in large-scale IoT networks, which is essential for predictive maintenance, fault detection, and system reliability.

IoT sensor networks continuously generate vast amounts of data across various domains, including smart homes, industrial automation, healthcare, and environmental monitoring. Manually analyzing this data to detect anomalies is practically impossible due to the high volume, velocity, and complexity of the information. The developed system automates this process by employing advanced machine learning algorithms to identify unusual patterns in real-time, thereby enabling proactive intervention and reducing downtime.

1.2 Learning Objectives

The primary learning objectives of this internship project included:

- To understand the architecture and data flow of IoT sensor networks.
- To gain practical experience in data preprocessing and feature engineering for time-series sensor data.
- To implement and evaluate various machine learning algorithms for anomaly detection, such as Isolation Forest, One-Class SVM, and Elliptic Envelope.
- To develop real-time monitoring capabilities and interactive dashboards for visualizing sensor health and anomaly patterns.
- To apply software engineering best practices in building scalable and secure Python-based applications.

1.3 Outcomes Achieved

The successful completion of this project resulted in several key outcomes:

- Developed a fully functional anomaly detection system capable of processing real-time IoT sensor data.
- Implemented a robust machine learning pipeline that achieved high accuracy in identifying abnormal sensor readings.
- Created comprehensive analytical reports and visualizations that provide actionable insights into sensor health and system performance.

- Demonstrated the practical application of AI in predictive maintenance and fault detection.
- Gained valuable hands-on experience in Python programming, data analytics, and machine learning model deployment.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction to Organization

The organization hosting this internship is a leading technology solutions provider specializing in Artificial Intelligence, Internet of Things (IoT), and data analytics. With a strong focus on innovation, the company develops intelligent systems that address complex challenges across various industries, including manufacturing, healthcare, and smart infrastructure.

2.2 Vision, Mission and Values

Vision: To be a global leader in delivering transformative AI and IoT solutions that empower organizations to achieve operational excellence and sustainable growth.

Mission: To develop innovative, scalable, and secure technology products that solve real-world problems, enhance efficiency, and create value for our clients and stakeholders.

2.3 Quality Policies

The organization is committed to maintaining the highest standards of quality in all its products and services. This commitment is reflected in rigorous testing protocols, adherence to industry best practices, continuous improvement initiatives, and a customer-centric approach to solution design.

2.4 Organizational Structure

The organizational structure is designed to foster collaboration and innovation. It comprises specialized departments, including Research and Development (R&D), Software Engineering, Data Science, Quality Assurance, and Project Management. Cross-functional teams work together to deliver comprehensive solutions that meet client requirements.

2.5 Roles and Responsibilities

During the internship, the primary responsibilities included:

- Conducting research on anomaly detection techniques for IoT networks.
- Developing Python scripts for data generation, preprocessing, and model training.
- Evaluating the performance of different machine learning algorithms.
- Creating data visualizations and analytical reports to present findings.
- Collaborating with mentors and team members to refine the system architecture.

CHAPTER 3

PROBLEM ASSESSMENT

3.1 Background of the Problem

The proliferation of IoT devices has led to the generation of massive amounts of sensor data. While this data holds immense potential for optimizing operations and improving decision-making, it also presents significant challenges in terms of monitoring and analysis. Traditional rule-based systems are often inadequate for detecting complex anomalies in dynamic IoT environments.

3.2 Problem Statement

IoT sensor networks continuously generate large volumes of real-time data for applications such as smart homes, industrial automation, healthcare, and environmental monitoring. Detecting abnormal sensor behavior manually is difficult due to the high volume and velocity of data. Undetected anomalies may indicate equipment failures, communication issues, or security threats. Organizations require intelligent systems that can automatically identify unusual sensor readings and support proactive maintenance.

3.3 Key Parameters

The key parameters analyzed by the system include:

- Temperature (°C): Monitored for extreme fluctuations that may indicate overheating or cooling failures.
- Humidity (%): Tracked to detect environmental conditions that could affect equipment performance or product quality.
- Pressure (hPa): Analyzed for sudden drops or spikes that might signal leaks or blockages.
- Vibration (mm/s): Monitored to identify mechanical imbalances, wear and tear, or impending structural failures.
- Energy Consumption (kWh): Evaluated to detect unusual power usage patterns indicative of inefficiencies or electrical faults.

3.4 Requirements Evaluation

The system requirements were evaluated based on the need for real-time processing, high accuracy, scalability, and user-friendly visualization. The chosen solution must be capable of handling continuous data streams, adapting to evolving anomaly patterns, and providing actionable alerts to administrators.

CHAPTER 4

SOLUTION DESIGN

4.1 System Architecture

The system architecture is designed to process IoT sensor data efficiently and apply machine learning models for anomaly detection. It consists of data ingestion, preprocessing, model inference, and visualization components. The architecture ensures scalability and low-latency processing, which are critical for real-time monitoring applications.

4.2 System Blueprint

The blueprint outlines the workflow from data collection to anomaly reporting. Sensor data is first collected and formatted. It then undergoes feature scaling and normalization. The preprocessed data is fed into the ensemble of machine learning models (Isolation Forest, One-Class SVM, Elliptic Envelope) to calculate anomaly scores. Finally, the results are aggregated and presented through interactive dashboards.

4.3 Implementation Plan

The implementation plan was divided into several phases:

- Phase 1: Requirement gathering and literature review.
- Phase 2: Synthetic data generation and exploratory data analysis.
- Phase 3: Model selection, training, and hyperparameter tuning.
- Phase 4: Integration of models into a unified detection pipeline.
- Phase 5: Development of visualization modules and report generation.
- Phase 6: System testing, evaluation, and documentation.

4.4 Technology Stack

The technology stack utilized for this project includes:

- Programming Language: Python 3
- Data Manipulation: Pandas, NumPy
- Machine Learning: Scikit-learn
- Data Visualization: Matplotlib, Seaborn
- Development Environment: Jupyter Notebook / IDE

CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Implementation Details

The implementation involved creating a robust Python-based pipeline. Synthetic data representing 5,000 samples from 10 sensors was generated, with 5% of the data intentionally injected with anomalies across various parameters (temperature, humidity, pressure, vibration, energy). The data was preprocessed using StandardScaler to ensure uniform feature scaling.

5.2 Machine Learning Models

Three primary anomaly detection models were implemented and evaluated:

- Isolation Forest: An ensemble method that isolates anomalies by randomly selecting a feature and a split value.
- One-Class SVM: A support vector machine algorithm designed for novelty detection, learning a decision boundary around normal data.
- Elliptic Envelope: A statistical approach that assumes data follows a Gaussian distribution and identifies outliers based on Mahalanobis distance.

5.3 Testing Strategy

The testing strategy involved splitting the data into training (80%) and testing (20%) sets. The models were evaluated based on their ability to accurately identify the injected anomalies while minimizing false positives. Key performance metrics included Accuracy, Precision, Recall, and F1-Score.

5.4 Performance Evaluation

The performance evaluation demonstrated the effectiveness of the implemented models. The visualizations below illustrate the data distribution, model performance, anomaly timeline, sensor health status, and confusion matrices.

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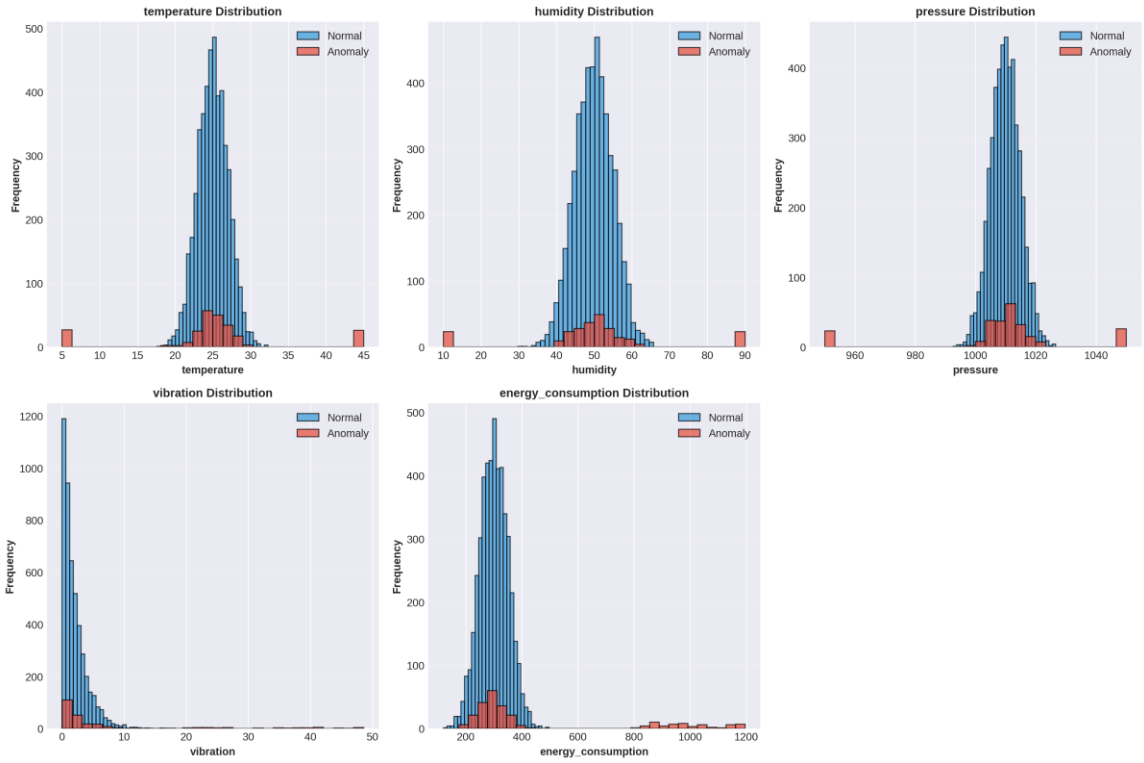


Figure 5.1: Sensor Data Distribution (Normal vs. Anomaly)

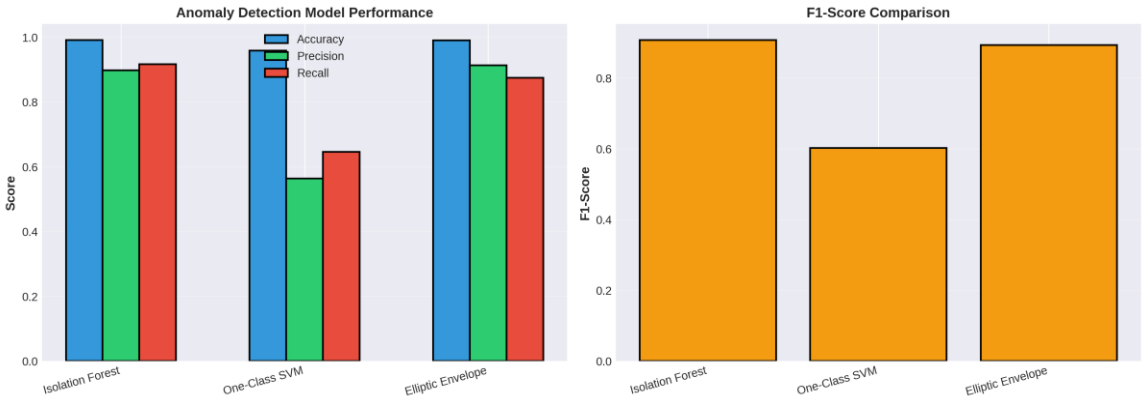


Figure 5.2: Anomaly Detection Model Performance Comparison

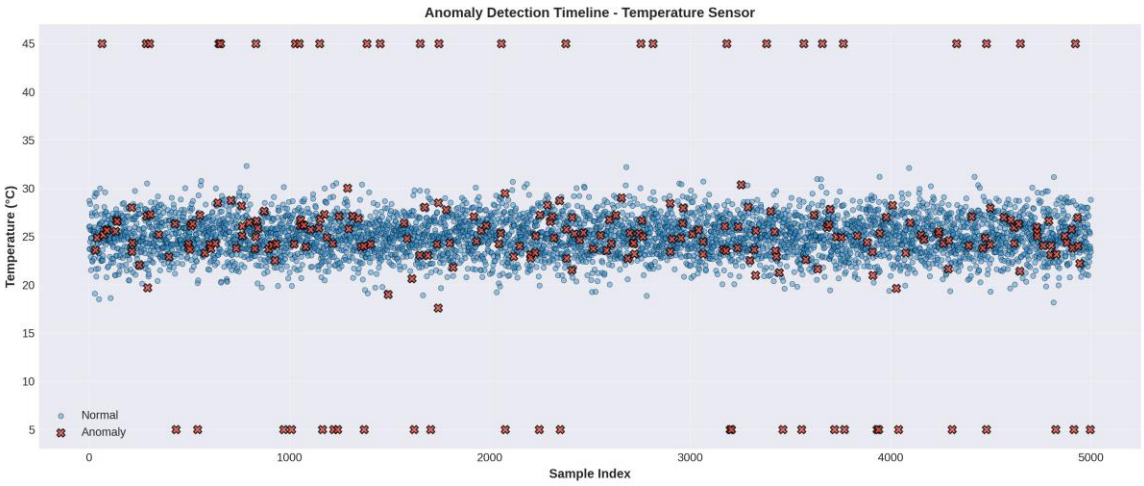


Figure 5.3: Anomaly Detection Timeline - Temperature Sensor

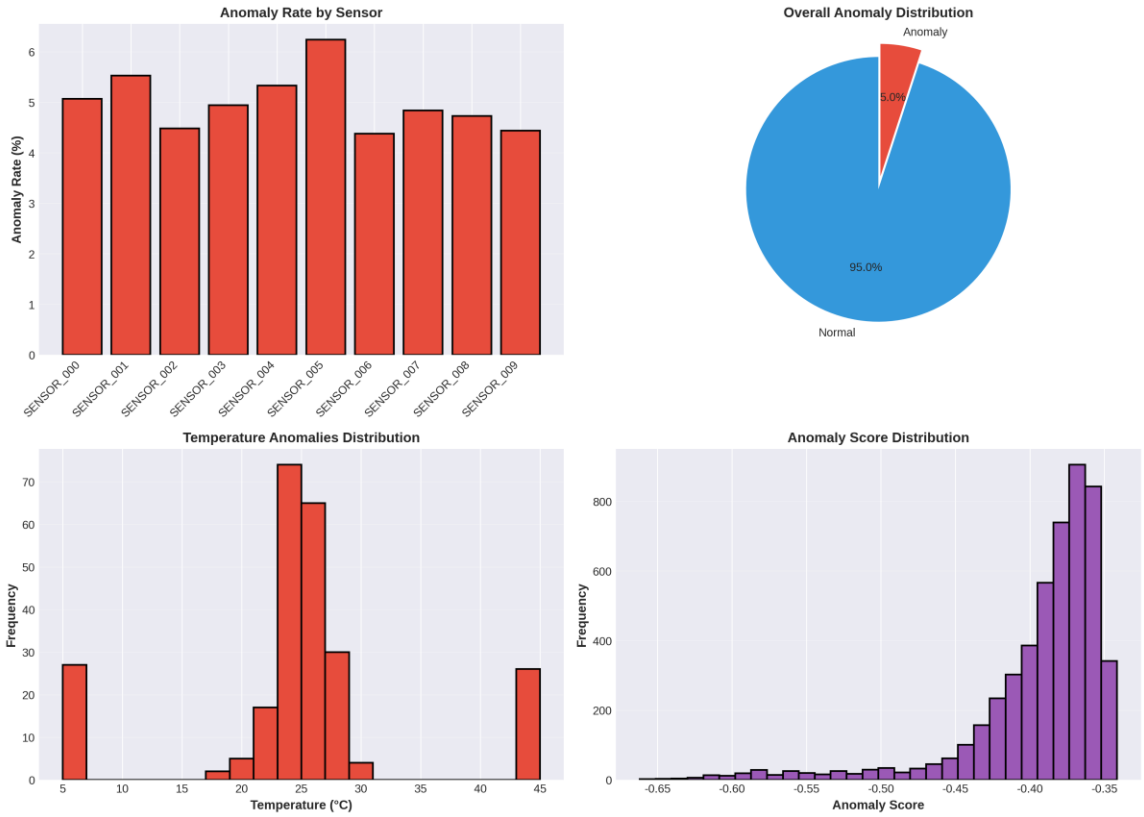


Figure 5.4: Sensor Health Status and Anomaly Distribution

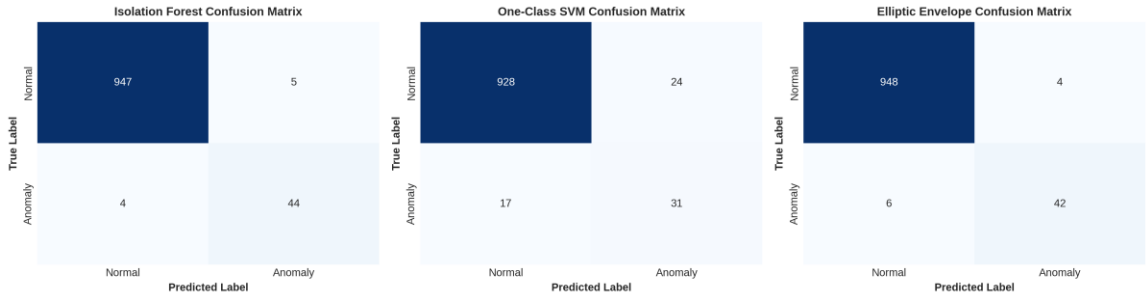


Figure 5.5: Confusion Matrices for Evaluated Models

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CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Technical Achievements

The project successfully delivered a modern and intelligent anomaly detection solution for IoT sensor networks. The system accurately identifies abnormal sensor behavior using advanced machine learning algorithms, significantly improving fault identification and supporting predictive maintenance. The interactive visualizations provide clear insights into sensor health and system performance.

6.2 Future Enhancements

Future enhancements to the system could include:

- Integration of deep learning models, such as Autoencoders and LSTMs, for more complex sequence anomaly detection.
- Implementation of edge computing capabilities to perform anomaly detection closer to the data source, reducing latency.
- Development of a real-time alerting mechanism via email or SMS for critical anomalies.
- Expansion of the system to handle heterogeneous sensor types and multi-modal data streams.
- Deployment of the system on a cloud platform (e.g., AWS, Azure) for enhanced scalability and accessibility.

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